Project Log (Notes)

07/09/2026

- Created ipynb file for exploring the data and plotting it
- Added to github repo

09/09/2026

- Within exploration notebook, used colab to help figure out how to extract the matlab file into useable arrays in python
- Then filtered to remove data entries where flank wear was NAN
- Plotted flank wear against time for every case, noticed that the wear rate seems to change between linear and exponential depending on settings
- Also noticed that case 6 is only a single entry with 0 wear at 0 seconds, should probably be removed from data analysis
- Noticed from the provided README that for each measurement, the mill was run for a few minutes, stopped to be measured, and the bit was reinserted repeatedly. This piecewise machining method may not be taking into account the heat that would be generated if ran repeatedly in industry

14/09/2026

- Should probably add some extra filtering even if I can see there's no duplicates
- Can use linear regression as a baseline, machine learning probably better

30/09/2026

- Removed all cases with only a single entry (Case 6 here)
- Created second notebook
- Created functions folder and file
- Created convert_data function to speed up code

07/10/2026

- Researched meaning of many milling terms (https://www.making.unsw.edu.au/unsw-cbr-makerspace/learning-resources/cnc-milling/cnc-feeds-speeds-milling/)
- Found RMS of all cutting vibrations and plotted with time
- Tried to understand how the time series measurements worked
- Regardless of time between measurements, each one had 9000 values
- From the README, sampling rate was 250Hz, meaning each section of 9000 values was ~36 seconds of data? Time values are not spaced like that, seems

like the vibrations aren't even from exactly the same time as they're grouped with
- Readme doesn't specify the units of time, I'm going to assume it must be minutes, only logical amount of time
- Going to neglect time series data, both doesn't make physical sense (Power drawn by machine will not affect physical wear rate) and is not something we would have access to before a run anyways, Feed, DOC, and material are concrete things that we (the engineer) can change before using the machine.
- Even if we fit a sensor to the machine for these variables mid-run, it would be incredibly complicated and out of scope to find a way to change other variables mid-cut in response, especially since they likely have a much smaller effect on wear rate compared to the physical parameters
- Separated data into training and validation, plotted which points are which

14/10/2026

- Fitted both a linear regression model and a neural network model, found them to both be quite inaccurate and bad at extrapolating
- Quite high RMSE on both linear and neural network
- Spoke with TA and was advised to use random training/validation split since the model cannot predict beyond the region of data it is given with the future validation split from notebook 5.6
- Realised that since we are not given any units for materials, should not be using binary 1 or 2 in lin-reg, since we need to know the physical units (Notebook 5.1). Could either neglect or add densities of the materials.
- Will add densities next and see if difference is made
- Spoke with Callum about how to handle the time series vibration data, he suggested trying many different things and looking for improvement, could maybe create a third notebook for exploring time series data
- He suggested trying peak magnitudes, average magnitudes, etc. in the neural network model

21/10/2026

- Realised that densities of cast iron and steel vary by type, cannot just research a number, and the README does not specify the exact type
- Introduced both average and peak magnitudes of the vibration and acoustic emission time series, finding that using the averages of the vibration values gives lowest validation RMSE
- Using more than two of any of the time series averages seems to give worse validation error, maybe because it starts to put too much importance on the time series over the other features?

22/10/2026

- Finished testing and created main code notebook
- Used copilot to assist writing functions to use model both for wear at a certain point, and for time reaching threshold wear

23/10/2026

- Added all functions to functions folder and created docstrings
- Added README and licence
- Added time series extension to the end of the folder

24/10/2026

- Finished up documentation and tests